



Derivatives Using a Table of Values

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Apply the sum, difference, product, and quotient rules to find derivatives of combined functions
- Evaluate derivatives at specific values using a given table of function and derivative values
- Interpret tabular data to compute $h'(x)$ for composite expressions involving $f(x)$ and $g(x)$

Use the table of values below to evaluate each derivative expression at the indicated point, showing the derivative rule applied in each step.

1. Given $h(x) = f(x) + 3g(x)$, find $h'(-2)$ using the table of values.

$$h(x) = f(x) + 3g(x), \quad h'(-2) = ?$$

Answer: _____

2. Given $h(x) = f(x) + g(x)$, find $h'(1)$ using the table of values.

$$h(x) = f(x) + g(x), \quad h'(1) = ?$$

Answer: _____

3. Given $h(x) = f(x) \cdot g(x)$, find $h'(0)$ using the product rule and the table of values.

$$h(x) = f(x) g(x), \quad h'(0) = ?$$

Answer: _____

4. Given $h(x) = \frac{f(x)}{g(x)}$, find $h'(2)$ using the quotient rule and the table of values.

$$h(x) = \frac{f(x)}{g(x)}, \quad h'(2) = ?$$

Answer: _____

5. Given $h(x) = 5f(x) - 2g(x)$, find $h'(-1)$ using the table of values.

$$h(x) = 5f(x) - 2g(x), \quad h'(-1) = ?$$

Answer: _____

6. Given $h(x) = f(x) \cdot g(x)$, find $h'(1)$ using the product rule and the table of values.

$$h(x) = f(x) g(x), \quad h'(1) = ?$$

Answer: _____



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7. Given $h(x) = \frac{g(x)}{f(x)}$, find $h'(-2)$ using the quotient rule and the table of values.

$$h(x) = \frac{g(x)}{f(x)}, \quad h'(-2) = ?$$

Answer: _____

8. Given $h(x) = 4f(x) + g(x)$, find $h'(2)$ using the table of values.

$$h(x) = 4f(x) + g(x), \quad h'(2) = ?$$

Answer: _____

9. Given $h(x) = f(x) \cdot g(x)$, find $h'(-1)$ using the product rule and the table of values.

$$h(x) = f(x) \cdot g(x), \quad h'(-1) = ?$$

Answer: _____

10. Given $h(x) = \frac{f(x)}{g(x)}$, find $h'(1)$ using the quotient rule and the table of values.

$$h(x) = \frac{f(x)}{g(x)}, \quad h'(1) = ?$$

Answer: _____





Provide students with the reference table: $x = -2, -1, 0, 1, 2$ with $f(x) = 3, 2, -1, 4, 0$; $f'(x) = 1, -2, 5, -3, 6$; $g(x) = -4, 1, 2, 5, 7$; $g'(x) = 8, 4, -1, 6, 2$.

Solutions

1. Given $h(x) = f(x) + 3g(x)$, find $h'(-2)$ using the table of values.

$$h(x) = f(x) + 3g(x), \quad h'(-2) = ?$$

- Differentiate using the sum and constant multiple rules to get $h'(x) = f'(x) + 3g'(x)$.
- Substitute $x = -2$ to obtain $h'(-2) = f'(-2) + 3g'(-2)$.
- From the table, $f'(-2) = 1$ and $g'(-2) = 8$.
- Compute $h'(-2) = 1 + 3(8) = 1 + 24 = 25$.

Answer: $h'(-2) = 25$

2. Given $h(x) = f(x) + g(x)$, find $h'(1)$ using the table of values.

$$h(x) = f(x) + g(x), \quad h'(1) = ?$$

- Apply the sum rule to get $h'(x) = f'(x) + g'(x)$.
- Substitute $x = 1$ to obtain $h'(1) = f'(1) + g'(1)$.
- From the table, $f'(1) = -3$ and $g'(1) = 6$.
- Compute $h'(1) = -3 + 6 = 3$.

Answer: $h'(1) = 3$

3. Given $h(x) = f(x) \cdot g(x)$, find $h'(0)$ using the product rule and the table of values.

$$h(x) = f(x)g(x), \quad h'(0) = ?$$

- Apply the product rule: $h'(x) = f'(x)g(x) + f(x)g'(x)$.
- Substitute $x = 0$ to get $h'(0) = f'(0)g(0) + f(0)g'(0)$.
- From the table, $f(0) = -1$, $f'(0) = 5$, $g(0) = 2$, and $g'(0) = -1$.
- Compute $h'(0) = (5)(2) + (-1)(-1) = 10 + 1 = 11$.

Answer: $h'(0) = 11$

4. Given $h(x) = \frac{f(x)}{g(x)}$, find $h'(2)$ using the quotient rule and the table of values.

$$h(x) = \frac{f(x)}{g(x)}, \quad h'(2) = ?$$

- Apply the quotient rule: $h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$.
- Substitute $x = 2$ to get $h'(2) = \frac{f'(2)g(2) - f(2)g'(2)}{(g(2))^2}$.
- From the table, $f(2) = 0$, $f'(2) = 6$, $g(2) = 7$, and $g'(2) = 2$.
- Compute $h'(2) = \frac{(6)(7) - (0)(2)}{(7)^2} = \frac{42}{49} = \frac{6}{7}$.

Answer: $h'(2) = \frac{42}{49} = \frac{6}{7}$



5. Given $h(x) = 5f(x) - 2g(x)$, find $h'(-1)$ using the table of values.

$$h(x) = 5f(x) - 2g(x), \quad h'(-1) = ?$$

→ Apply constant multiple and difference rules to get $h'(x) = 5f'(x) - 2g'(x)$.

→ Substitute $x = -1$ to obtain $h'(-1) = 5f'(-1) - 2g'(-1)$.

→ From the table, $f'(-1) = -2$ and $g'(-1) = 4$.

→ Compute $h'(-1) = 5(-2) - 2(4) = -10 - 8 = -18$.

Answer: $h'(-1) = -18$

6. Given $h(x) = f(x) \cdot g(x)$, find $h'(1)$ using the product rule and the table of values.

$$h(x) = f(x)g(x), \quad h'(1) = ?$$

→ Apply the product rule: $h'(x) = f'(x)g(x) + f(x)g'(x)$.

→ Substitute $x = 1$ to get $h'(1) = f'(1)g(1) + f(1)g'(1)$.

→ From the table, $f(1) = 4$, $f'(1) = -3$, $g(1) = 5$, and $g'(1) = 6$.

→ Compute $h'(1) = (-3)(5) + (4)(6) = -15 + 24 = 9$.

Answer: $h'(1) = 9$

7. Given $h(x) = \frac{g(x)}{f(x)}$, find $h'(-2)$ using the quotient rule and the table of values.

$$h(x) = \frac{g(x)}{f(x)}, \quad h'(-2) = ?$$

→ Apply the quotient rule: $h'(x) = (g'(x)f(x) - g(x)f'(x)) / (f(x))^2$.

→ Substitute $x = -2$ to get $h'(-2) = (g'(-2)f(-2) - g(-2)f'(-2)) / (f(-2))^2$.

→ From the table, $f(-2) = 3$, $f'(-2) = 1$, $g(-2) = -4$, and $g'(-2) = 8$.

→ Compute $h'(-2) = ((8)(3) - (-4)(1)) / (3)^2 = (24 + 4) / 9 = 28/9$.

Answer: $h'(-2) = \frac{28}{9}$

8. Given $h(x) = 4f(x) + g(x)$, find $h'(2)$ using the table of values.

$$h(x) = 4f(x) + g(x), \quad h'(2) = ?$$

→ Apply constant multiple and sum rules to get $h'(x) = 4f'(x) + g'(x)$.

→ Substitute $x = 2$ to obtain $h'(2) = 4f'(2) + g'(2)$.

→ From the table, $f'(2) = 6$ and $g'(2) = 2$.

→ Compute $h'(2) = 4(6) + 2 = 24 + 2 = 26$.

Answer: $h'(2) = 26$

9. Given $h(x) = f(x) \cdot g(x)$, find $h'(-1)$ using the product rule and the table of values.

$$h(x) = f(x)g(x), \quad h'(-1) = ?$$

→ Apply the product rule: $h'(x) = f'(x)g(x) + f(x)g'(x)$.

→ Substitute $x = -1$ to get $h'(-1) = f'(-1)g(-1) + f(-1)g'(-1)$.

→ From the table, $f(-1) = 2$, $f'(-1) = -2$, $g(-1) = 1$, and $g'(-1) = 4$.

→ Compute $h'(-1) = (-2)(1) + (2)(4) = -2 + 8 = 6$.

Answer: $h'(-1) = 6$



10. Given $h(x) = \frac{f(x)}{g(x)}$, find $h'(1)$ using the quotient rule and the table of values.

$$h(x) = \frac{f(x)}{g(x)}, \quad h'(1) = ?$$

→ Apply the quotient rule: $h'(x) = (f'(x)g(x) - f(x)g'(x)) / (g(x))^2$.

→ Substitute $x = 1$ to get $h'(1) = (f'(1)g(1) - f(1)g'(1)) / (g(1))^2$.

→ From the table, $f(1) = 4$, $f'(1) = -3$, $g(1) = 5$, and $g'(1) = 6$.

→ Compute $h'(1) = ((-3)(5) - (4)(6)) / (5)^2 = (-15 - 24) / 25 = -39/25$.

Answer: $h'(1) = -\frac{39}{25}$

